Notes for Distributed Random Sampling

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Introduction

Sampling from a Dataset: Centralized Algorithms (1/2)

The first task we will study is generating and maintaining a **fixed-size** *sampling*, without replacement, from a dataset A of n elements. In particular, we will show the following results:

1. **Uniform Random Sample from a Data Array.** Given a set A of n elements stored in an array $\vec{A} = \vec{A}[1:n]$, the *Fisher-Yates Shuffle* algorithm iteratively generates a u.a.r. permutation $\vec{A}_\pi$ of $\vec{A}$. If we define $\mathcal{S}$ as the first s elements of $\vec{A}_\pi$, we will show that $\mathcal{S}$ satisfies:
    - **a).** As an unordered subset, $\mathcal{S}$ is uniformly distributed over the space $\binom{n}{s}$ of all subsets of size s of A;
    - **b).** Moreover, $\vec{\mathcal{S}} = \vec{\mathcal{S}}[1:s]$, as an ordered set, is ordered u.a.r. relative to all $s!$ possible orderings.

This part is covered in Section **??**.

Sampling from a Dataset: Centralized Algorithms (2/2)

2. **Uniform Random Sample from a Data Stream.** Consider a model where input data arrives at a server C in *streaming* mode: $a_1, a_2, \ldots, a_k, \ldots$. The task for C is to generate and maintain *on-line* a sampling $\mathcal{S}(k)$ of size s without replacement with these properties:
   - **a1).** At the end of round k, having read the first k elements on-line, the output $\mathcal{S}(k)$ (unordered) must be uniformly distributed over all subsets of size s of $\vec{\mathcal{A}}[1, k]$;
   - **b1).** The algorithm must maintain a random permutation of $\mathcal{S}(k)$.

This part is covered in Section 3.

Distributed Sampling

Merging Distributed Random Samplings. Consider a star-shaped distributed computing model with a central server C and $m \geq 2$ peripheral nodes $P_1, \ldots, P_m$.

The central node C must compute a sampling $\mathcal{S}$ of the global dataset A partitioned into m disjoint subsets $A_1, \ldots, A_m$, each stored privately by a peripheral node. C has no direct access to this data.

The task is to generate local samplings $\mathcal{S}_1, \ldots, \mathcal{S}_m$ and allow C to generate $\mathcal{S}$ by communicating with the nodes, optimizing **message complexity**:

- **Naive Protocol:** Nodes send samples and local sizes n_i. Complexity: $\tilde{\Theta}(m \cdot s)$.
- **Optimal Protocol:** Uses the *Principle of Deferred Decisions*. Complexity: $\tilde{\Theta}(m + s)$.

Permutations and Sampling

Fisher-Yates Shuffle (FYA)

Algorithm FYA

Input: $\vec{\mathcal{A}}[1:n]$ of n distinct integers.

Output: A u.a.r. permutation $\vec{\mathcal{A}}_\pi$ of $\vec{\mathcal{A}}$.

1. **Set** $\vec{\mathcal{A}}_\pi[1:n] := A[1:n]$
2. **For** $i = 1$ **To** n **Do**
    - **Choose** $j \in_u [i]$;
    - **Swap** $\vec{\mathcal{A}}_\pi[i]$ with $\vec{\mathcal{A}}_\pi[j]$.
3. **Return** $\vec{\mathcal{A}}_\pi[1:n]$

Proof of Lemma 1 (1/2)

Lemma

At the end of each round $i = 1, \ldots, n$, the prefix $\vec{\mathcal{A}}_\pi[1 : i]$ is a random permutation of the first i elements of the initial vector $A[1 : n]$.

Proof: By induction on i. Let $\langle a_1, \ldots, a_i \rangle$ be any sub-permutation of $A[1 : i]$. We calculate $\mathbb{P}\left(\mathcal{E}(i)\right)$ where $\mathcal{E}(i) =$ "$\vec{\mathcal{A}}_\pi[1 : i] = \langle a_1, \ldots, a_i \rangle$".

- **For $i = 1$:** $\mathbb{P}\left(\mathcal{E}(1)\right) = 1/1! = 1$. Trivially true.
- **Inductive Assumption:** For $i \geq 2$, at the end of round $i - 1$, $\mathbb{P}\left(\mathcal{E}(i-1)\right) = 1/(i-1)!$.

Proof of Lemma 1 (2/2)

Only two mutually exclusive cases can occur based on the choice $j = J(i)$:

- **Case a).** $J(i) = i$. The only favorable configuration for $\mathcal{E}(i)$ was that $\vec{\mathcal{A}}[i] = a_i$ initially. Since $\mathbb{P}\left(J(i) = i\right) = 1/i$, by independence and the inductive hypothesis:

$$\mathbb{P}\left(\mathcal{E}(i)\right) = \frac{1}{(i-1)!} \cdot \frac{1}{i}$$

- **Case b).** $J(i) = k$ for some $k < i$. Let $a_i = x$. The only favorable configuration was that at round $i - 1$:

$$\vec{\mathcal{A}}_\pi[1, (i-1)] = \langle a_1, \ldots, a_k = x, \ldots, a_{i-1} \rangle$$

  This configuration is unique, allowing the same argument as case (a).

$\square$

Uniform Sampling of Fixed Size

Algorithm FY-Sample

Input: Array $\vec{\mathcal{A}}[1:n]$ of distinct numbers; $s \in [n]$

Output: A sampling $\mathcal{S} \subseteq A$

1. **Set** $\mathcal{S} := \emptyset$
2. **Execute** $\mathrm{FYA}[\vec{\mathcal{A}}[1:n]]$
3. **Return** $\mathcal{S} := \vec{\mathcal{A}}_\pi[1:s]$.

Lemma

The distribution of $\mathcal{S}$ satisfies: **a).** *As an unordered subset, $\mathcal{S}$ is uniform over $\binom{n}{s}$.* **b).** *$\vec{\mathcal{S}}[1:s]$ is ordered u.a.r. relative to all $s!$ orderings.*

Proof: $\mathbb{P}\left(\mathcal{S} = S\right) = \frac{s!(n-s)!}{n!} = \frac{1}{\binom{n}{s}}$.

Sampling from a Data Stream

Sampling from a Data Stream: The Model

In the previous section, we considered a centralized framework where algorithms have access to a central memory where the entire data vector is present from the beginning.

In this section, instead, we consider the classic *Streaming* model in which the algorithm receives as input a (potentially infinite) sequence (stream) of data:

$$\textsc{Streaming Input:} \quad a_1, \ldots, a_k, \ldots, \tag{1}$$

More precisely, at round $k \geq 1$, the algorithm receives the element a_k and must process and make decisions based on what it has read (and possibly stored) from the subsequence $a_1, \ldots, a_k$.

The One-Passing Model

The model is *one-passing*: it is not possible to retrieve a past element of the stream unless it has been stored in the internal memory of the processor.

Typically, the goal is to use as little memory space as possible to answer certain fixed queries on the stream in the shortest possible time (*query time*).

Problem P2: Online Sampling

The task in the aforementioned online model that we analyze in this section is the following:

Problem P2. *Given the Streaming Input 1, generate and maintain, at each round* $k \geq 1$*, a sampling of size* s*, distributed u.a.r.,* $\mathcal{S}(s, k)$ *of the input set of elements* $a_1, \ldots, a_k$*.*

Uniformity Objective

The objective consists in maintaining, at each round $k \geq s$, a random subset $\mathcal{S}(s,k)$ of size exactly s (typically $s \ll n$), such that:

$$\forall S' \in \binom{k}{s}, \quad \mathbb{P}(\mathcal{S}(s,k) = S') = \frac{1}{\binom{k}{s}} = \frac{s!(k-s)!}{k!} \tag{2}$$

Reservoir Sampling Algorithm (RSA)

We next describe the streaming algorithm that, at every round k, receives element a_k.

RSA Algorithm

Input: Stream $a_1, \ldots, a_k, \ldots$; $s \in [n]$

1. **Set** $\mathcal{S}(s,k) := \emptyset$;
2. **For Each** $k \leq s$, **Do** $\mathcal{S}(s,k) := \mathcal{S}(s,k) \cup \{a_k\}$;
3. **For** $k = s+1, \ldots,$ **Do**
   - **Choose** $j \in_u [k]$;
   - **If** $j \leq s$ **Then Replace** a_j with a_k (Case i);
   - **If** $j \geq s+1$ **Then Discard** a_k (Case ii).
4. **Maintain and Return** $\mathcal{S}(s,k)$.

THEOREM 1. For any $s \geq 1$, at round $k \geq s$, for any $S' \subseteq \{a_1, \ldots, a_k\}$ with $|S'| = s$, the probability that RSA returns S' is $1/\binom{k}{s}$.

Proof of THM (1/4)

The proof is by induction on k, where the base case is $k = s$.

Induction Base ($k = s$): As long as $k \leq s$, RSA inserts the first k input elements into $\mathcal{S}(s, k)$. There is only one possible subset of size k, which is $\langle a_1, \ldots, a_k \rangle$ itself. The probability that $\mathcal{S}(s, k)$ coincides with this subset is $1/\binom{k}{k} = 1$. The base case is verified.

Inductive Step: Assume, by inductive hypothesis, that after processing k elements, every subset of size s of the elements $\{a_1, \ldots, a_k\}$ has probability $1/\binom{k}{s}$ of being chosen as the output $\mathcal{S}(s, k)$.

Now consider the element a_{k+1}. RSA generates a random index $j \in_u [1, k+1]$, and we have two disjoint and exhaustive events that determine the new sampling $\mathcal{S}$:

a). If $j > s$, the element a_{k+1} is discarded.
b). If $j \leq s$, the element a_{k+1} replaces the element at position j in $\mathcal{S}(s, k+1)$.

Proof of THM (2/4)

We must calculate the probability that a generic subset $S' \subseteq \{a_1, \ldots, a_{k+1}\}$ of size s is the new sample $\mathcal{S}(s, k+1)$. We distinguish the analysis into two distinct cases.

Case (a): $a_{k+1} \notin S'$. In this case, to have $\mathcal{S}(s, k+1) = S'$, S' must have been chosen from the first k elements: this occurs *if and only if* we have the conjunction of two distinct and independent events:

a.1). The element a_{k+1} must have been discarded. This happens if the extracted index j is $> s$. The probability is $\frac{(k+1)-s}{k+1}$; AND

a.2). S' was chosen *in the previous round* as $\mathcal{S}(s, k)$. This sub-event has probability $\frac{1}{\binom{k}{s}}$ by inductive hypothesis.

Therefore:

$$\mathbb{P}\left(\mathcal{S}(s, k+1) = S'\right) = \frac{k+1-s}{k+1} \cdot \frac{1}{\binom{k}{s}} = \frac{k+1-s}{k+1} \cdot \frac{s!(k-s)!}{k!} = \frac{1}{\binom{k+1}{s}}$$

Proof of THM (3/4)

Case (b): $a_{k+1} \in S'$. In this case, the previous sample $\mathcal{S}(s,k)$ must have contained the remaining $s-1$ elements of S' (which are *fixed*!), plus another element X (*any*!) which is replaced by a_{k+1} at round $k+1$.

Define S_{fixed} as the set of the other $s-1$ elements composing S':

$$S_{fixed} = S' - \{a_{k+1}\}$$

The old sample $\mathcal{S}(s,k)$ must have been: $\mathcal{S}_k = S_{fixed} \cup \{X\}$, where $X \in \{a_1, \ldots, a_k\} - S_{fixed}$.

The number of ways to choose X (the element occupying the "removed spot") is:

$$\text{n. of choices for } X = k - (s-1) = k - s + 1$$

So, there are exactly $(k-s+1)$ different subsets $\mathcal{S}(s,k)$ that can transform into S' via the replacement of X with a_{k+1}.

Proof of THM (4/4)

Since these configurations are mutually exclusive, the total probability is the sum:

$$\mathbb{P}\left(\mathcal{S}(s,k+1)=S'\right)=\sum_{x\in\{a_1,\ldots,a_k\}\setminus S_{fixed}}\mathbb{P}\left(\mathcal{S}(s,k)=S_{fixed}\cup\{x\}\right)\cdot\mathbb{P}\left(a_{k+1}\text{ replaces } x\right)$$

By inductive hypothesis, $\mathbb{P}\left(\mathcal{S}(s,k)=S_k\right)=1/\binom{k}{s}$. Moreover:

- Probability a_{k+1} enters is $\frac{s}{k+1}$.
- Probability it replaces specifically x (and not the others) is $\frac{1}{s}$.

$$\mathbb{P}\left(\mathcal{S}(s,k+1)=S'\right)=(k-s+1)\cdot\left[\frac{1}{\binom{k}{s}}\cdot\frac{s}{k+1}\cdot\frac{1}{s}\right]$$

Developing the binomial coefficient:

$$\begin{aligned}\mathbb{P}\left(\mathcal{S}(s,k+1)=S'\right)&=(k-s+1)\cdot\frac{s!(k-s)!}{k!}\cdot\frac{1}{k+1}\\&=\frac{s!(k-s+1)!}{(k+1)!}=\frac{1}{\binom{k+1}{s}}\end{aligned}$$

This completes the proof. □

Main Reference

Jeffrey S. Vitter (1985), *"Random Sampling with a Reservoir"*.

Ordered Sampling from Streams

We saw how to generate a random ordering of a vector $A[1:n]$ using the FYA algorithm, and we presented the RSA algorithm which generates and maintains a fixed-size sample from a stream of elements $a_1, \ldots, a_k, \ldots$.

We now aim to combine these two algorithms to efficiently maintain "online" a sampling that possesses both the above properties. We seek an online algorithm to solve the following:

Problem P3. *Given the Streaming Input*

$$\textsc{Streaming Input:} \quad a_1, \ldots, a_k, \ldots, \tag{3}$$

generate and maintain, at each round $k \geq 1$, an ordered sampling $\vec{\mathcal{S}}(s,k)$ of size s such that:

a) *It is chosen u.a.r. among all possible subsets of size s of $\{a_1, \ldots, a_k\}$;*
b) *Its elements are ordered u.a.r.*

PRSA: Permuting Reservoir Sampling Algorithm

Input: Input Stream (3); $s \in [n]$

Set $\vec{\mathcal{S}}(s,0)[1:s] := \vec{0}$;

For Each $1 \leq k \leq s$ **Do:**

Set $\vec{\mathcal{S}}(s,k)[k] := a_k$; **Choose** $i \in_u [k]$;

Swap $\vec{\mathcal{S}}(s,k)[i]$ with $\vec{\mathcal{S}}(s,k)[k]$. $*$ Apply FYA $*$

For $k = s+1, \ldots,$ **Do** $*$ Apply the two cases of RSA $*$

Choose $j \in_u [k]$

If $j \leq s$ **Then Set** $\vec{\mathcal{S}}(s,k)[1:s]$ as follows: $**$ Case (i) $**$

- **Remove** a_j from $\vec{\mathcal{S}}(s,k-1)[1:s]$;
- The first $s-1$ elements of $\vec{\mathcal{S}}(s,k)[1:s]$ are as in $\vec{\mathcal{S}}(s,k-1)[1:s]$;
- **Set** $\vec{\mathcal{S}}(s,k)[s] := a_k$; **Choose** $h \in_u [k]$
- **Swap** $\vec{\mathcal{S}}(s,k)[s]$ with $\vec{\mathcal{S}}(s,k)[h]$ $**$ Apply FYA $**$

If $j \geq s+1$ **Then:**

- **Discard** a_k; **Set** $\vec{\mathcal{S}}(s,k)[1:s] := \vec{\mathcal{S}}(s,k-1)[1:s]$. $**$ Case (ii) $**$

Maintain and Return $\mathcal{S}(s,k)[1:s]$.

Correctness and Complexity of PRSA

THEOREM 2. The algorithm PRSA solves Problem P3 in space and time $\tilde{O}(s)$ (per round).

Note on space: Although the procedure is described using distinct copies of the vector for clarity, all operations can be performed in-place using a single array of size s.

Sketch of Proof. The fact that $\vec{\mathcal{S}}(s, k)[1 : s]$ satisfies Properties (a) and (b) is obtained by applying Theorem 1 and Lemma 2.

- Thanks to the RSA strategy, the set $\mathcal{S}(s, k)$ is uniform over $\binom{k}{s}$.
- Regarding the ordering, we use induction on k.

Sketch of Proof (Continued)

- Assume $\vec{\mathcal{S}}(s, k-1)[1:s]$ is ordered u.a.r.
- **Conditioning on Case (i):** If a_k enters the reservoir, the sub-vector $\vec{\mathcal{S}}(s, k)[1:s-1]$ is, by construction, a u.a.r. *sub-permutation* of the $s-1$ surviving elements.
- By inserting a_k into a u.a.r. position $h \in_u [s]$, the analysis of the FYA Algorithm shows that the final vector $\vec{\mathcal{S}}(s, k)[1:s]$ is also ordered u.a.r.
- The case where a_k is discarded is trivial, as the random ordering is preserved from the previous round.

$\square$

Distributed Sampling: Merging Protocol

Distributed Merging of Random Samplings

Consider a **Star Network**: central server C and $h \geq 2$ peripheral nodes $P_1, \ldots, P_h$. The communication graph is $G = (V, E)$ where:

$$V = \{C, P_1, \ldots, P_h\} \text{ and } E = \{\{C, P_i\} : \ i = 1, \ldots, h\}$$

Problem P4
C must maintain a u.a.r. sample $\mathcal{S}$ of a global dataset D partitioned into h disjoint subsets $D_1, \ldots, D_h$, each managed privately by a node P_i. C has no direct access to peripheral data.

Trivial Idea: C asks all the data sets from nodes P_i and then sample s random elements from the union of them.

Basic Idea: Nodes generate local samples $\mathcal{S}_i$. C determines (via a local random process) the number s_i of elements to request from each P_i to form the global sample $\mathcal{S}$.

Preliminaries: Hypergeometric Distribution

The *Hypergeometric* distribution is the fundamental model for sampling **without replacement**.

Definition (Hypergeometric Distribution (n, m, s))
Given a dataset D of n elements with m "successes" (Type 1) and $n - m$ "failures" (Type 2), the number of successes X in a u.a.r. sample of size s follows $X \sim$ **Hypergeometric**(n, m, s).

Probability Mass Function (PMF): For $k \in \{\max(0, s - (n - m)), \ldots, \min(s, m)\}$:

$$P(X = k) = \frac{\binom{m}{k}\binom{n-m}{s-k}}{\binom{n}{s}}$$

Properties and Urn Interpretation

Fact
For $X \sim$ **Hypergeometric**(n, m, s)*:*

- $\mathbb{E}[X] = s \cdot \frac{m}{n}$
- $\mathrm{Var}(X) = s \cdot \frac{m}{n} \cdot \frac{n-m}{n} \cdot \frac{n-s}{n-1}$

The term $\frac{n-s}{n-1}$ *is the finite population correction factor.*

Urn Process: If $Y_1, \ldots, Y_s$ are binary variables ($Y_i = 1$ if i-th draw is success):

i) $\mathbb{P}(Y_1 = 1) = m/n$;
ii) $\mathbb{P}(Y_2 = 1 | Y_1 = 1) = \frac{m-1}{n-1}$;
iii) $\mathbb{P}(Y_2 = 1 | Y_1 = 0) = \frac{m}{n-1}$.

Exchangeability

Fact (Exchangeability)

The sequence $Y_1, \dots, Y_n$ is exchangeable. The probability of a sequence depends only on the total number of successes k:

$$P(Y_1 = y_1 \wedge \dots \wedge Y_s = y_s) = \frac{m!(n-m)!(n-s)!}{n!(m-k)!(n-m-(s-k))!}$$

Note: For $h \geq 3$ (multivariate case), the logic is similar but calculations are more complex. We focus on $h = 2$ (nodes P_1, P_2).

Weighted Hypergeometric Merging

Let $D = D_1 \cup D_2$ where D_1 (size n_1) is on P_1 and D_2 (size $\ell = n - m$) is on P_2.

Protocol whs

1. C requests $|D_1|$ and $|D_2|$ from P_1, P_2
2. C sets $n_1 := |D_1|$, $n_2 := |D_2|$ $n := n_1 + n_2$, and generates $k \sim \text{Hypergeometric}(n, n_1, s)$
3. C sends k to P_1 and $s - k$ to P_2
4. P_i extracts a sample $\mathcal{S}_i$ from D_i of size s, $i = 1, 2$
5. P_1 extracts a u.a.r. subset $\mathcal{R}_1$ from its sample $\mathcal{S}_1$ of size k. P_2 extracts $\mathcal{R}_1$ from $\mathcal{S}_B$ of size $s - k$. They send them to C
6. C maintains $\mathcal{S} := \mathcal{R}_1 \cup \mathcal{R}_2$

Main Theorem: Correctness of Merging

Theorem

Assuming the star system $G(V, E)$ and dataset $D = D_1 \cup D_2$, the protocol WHS *generates a random sample $\mathcal{S}$ uniformly at random (u.a.r.) of size s. The message complexity is $O(s + h)$, with $h = 2$.*

Proof Strategy: We must show that for any $X \subset D$ of size s:

$$P(\mathcal{S} = X) = \frac{1}{\binom{n}{s}}$$

Proof of Theorem 6 (1/3)

Let X contain k elements from D_1 and $s - k$ elements from D_2. The probability $P(\mathcal{S} = X)$ is the product of three events:

1. E_1**:** The Hypergeometric rnd variable K equals k.

$$P(E_1) = \frac{\binom{n_1}{k}\binom{n_2}{s-k}}{\binom{n}{s}}$$

2. E_2**:** $\mathcal{S}_1$ contains $X \cap D_1$ **AND** these are selected as $\mathcal{R}_1$ in Step 5.
3. E_3**:** $\mathcal{S}_2$ contains $X \cap D_2$ **AND** these are selected as $\mathcal{R}_2$ in Step 5.

Proof of Theorem 6 (2/3)

The probability that a specific set of k elements is in sample $\mathcal{S}_1$ of size s from n_1 is $\binom{n_1-k}{s-k}/\binom{n_1}{s}$. The probability of picking them from the sample is $1/\binom{s}{k}$.

Thus:

$$P(E_2) = \frac{\binom{n_1-k}{s-k}}{\binom{n_1}{s}} \cdot \frac{1}{\binom{s}{k}}$$

Detailed Calculation of $\mathbb{P}(E_3)$

Recall that E_3 is the event: "The specific set $X_2 \subset D_2$ (of size $s-k$) is included in the global sample R via node P_2."

This event requires two steps:

1. **Inclusion:** X_2 must be present in the local sample $\mathcal{S}_2$ (size s).
2. **Selection:** X_2 must be selected from $\mathcal{S}_2$ in Step 5 (where $s-k$ elements are picked and communicate to C).

1. Inclusion Probability: The number of ways to form $\mathcal{S}_2$ containing X_2 is $\binom{n_2-|X_2|}{s-|X_2|}$. Since $|X_2| = s-k$:

$$\mathbb{P}(X_2 \subset \mathcal{S}_2) = \frac{\binom{n_2-(s-k)}{s-(s-k)}}{\binom{n_2}{s}} = \frac{\binom{n_2-s+k}{k}}{\binom{n_2}{s}}$$

2. Selection Probability: Once X_2 is in $\mathcal{S}_2$, the probability of picking exactly those $s-k$ elements out of s available is:

$$\mathbb{P}(\text{pick } X_2 | X_2 \subset \mathcal{S}_2) = \frac{1}{\binom{s}{s-k}}$$

Step-by-Step Algebraic Simplification of $\mathbb{P}(E_3)$

Multiplying the two steps:

$$\mathbb{P}(E_3) = \underbrace{\frac{\binom{n_2-s+k}{k}}{\binom{n_2}{s}}}_{\text{Inclusion}} \cdot \underbrace{\frac{1}{\binom{s}{s-k}}}_{\text{Selection}}$$

Now, we use the **Committee Identity** on the denominator $\binom{n_2}{s}\binom{s}{s-k}$:

$$\binom{N}{K}\binom{K}{r} = \binom{N}{r}\binom{N-r}{K-r}$$

Setting $N = n_2$, $K = s$, and $r = s - k$:

$$\binom{n_2}{s}\binom{s}{s-k} = \binom{n_2}{s-k}\binom{n_2-(s-k)}{s-(s-k)} = \binom{n_2}{s-k}\binom{n_2-s+k}{k}$$

Substitute this back into the formula for $\mathbb{P}(E_3)$:

$$\mathbb{P}(E_3) = \frac{\binom{n_2-s+k}{k}}{\binom{n_2}{s-k}\binom{n_2-s+k}{k}} = \frac{1}{\binom{n_2}{s-k}}$$

Final Integration with the Hypergeometric Term

When we multiply $\mathbb{P}(E_1) \cdot \mathbb{P}(E_2) \cdot \mathbb{P}(E_3)$, look at what happens to the n_2 terms:

- From $\mathbb{P}(E_1)$ (Hypergeometric): We have $\binom{n_2}{s-k}$ in the **numerator**.
- From $\mathbb{P}(E_3)$ (Calculated just now): We have $\binom{n_2}{s-k}$ in the **denominator**.

$$\cdots \left[\frac{\text{Numerator}}{\binom{n}{s}} \cdot \frac{\binom{n_2}{s-k}}{1} \right] \cdots \cdots \left[\frac{1}{\binom{n_2}{s-k}} \right]$$

The term $\binom{n_2}{s-k}$ cancels out perfectly.

Summary

The complex binomial $\binom{n_2-s+k}{k}$ from the local inclusion probability is exactly what is needed to "transform" the local choice $\binom{n_2}{s}$ into the hypergeometric component $\binom{n_2}{s-k}$.

Proof of Theorem 6 (3/3)

Total probability $P(\mathcal{S} = X) = P(E_1) \cdot P(E_2) \cdot P(E_3)$:

$$P(\mathcal{S} = X) = \left[\frac{\binom{n_1}{k}\binom{n_2}{s-k}}{\binom{n}{s}} \right] \cdot \left[\frac{\binom{n_1-k}{s-k}}{\binom{n_1}{s}\binom{s}{k}} \right] \cdot \left[\frac{1}{\binom{n_2}{s-k}} \right]$$

Using $\binom{n}{k}\binom{k}{r} = \binom{n}{r}\binom{n-r}{k-r}$, we observe that: $\binom{n_1}{k}\binom{n_1-k}{s-k} = \binom{n_1}{s}\binom{s}{k}$.

All terms relating to n_1, n_2 and s cancel out perfectly:

$$P(\mathcal{S} = X) = \frac{1}{\binom{n}{s}}$$

The sample is a Simple Random Sample (SRS). $\square$